

NAME: S. ROHINI

REG. NO: 192424413

CODE: CSA0406

COURSE: OPERATING SYSTEMS

CO1 - AT2 $\Rightarrow$ CONCEPT

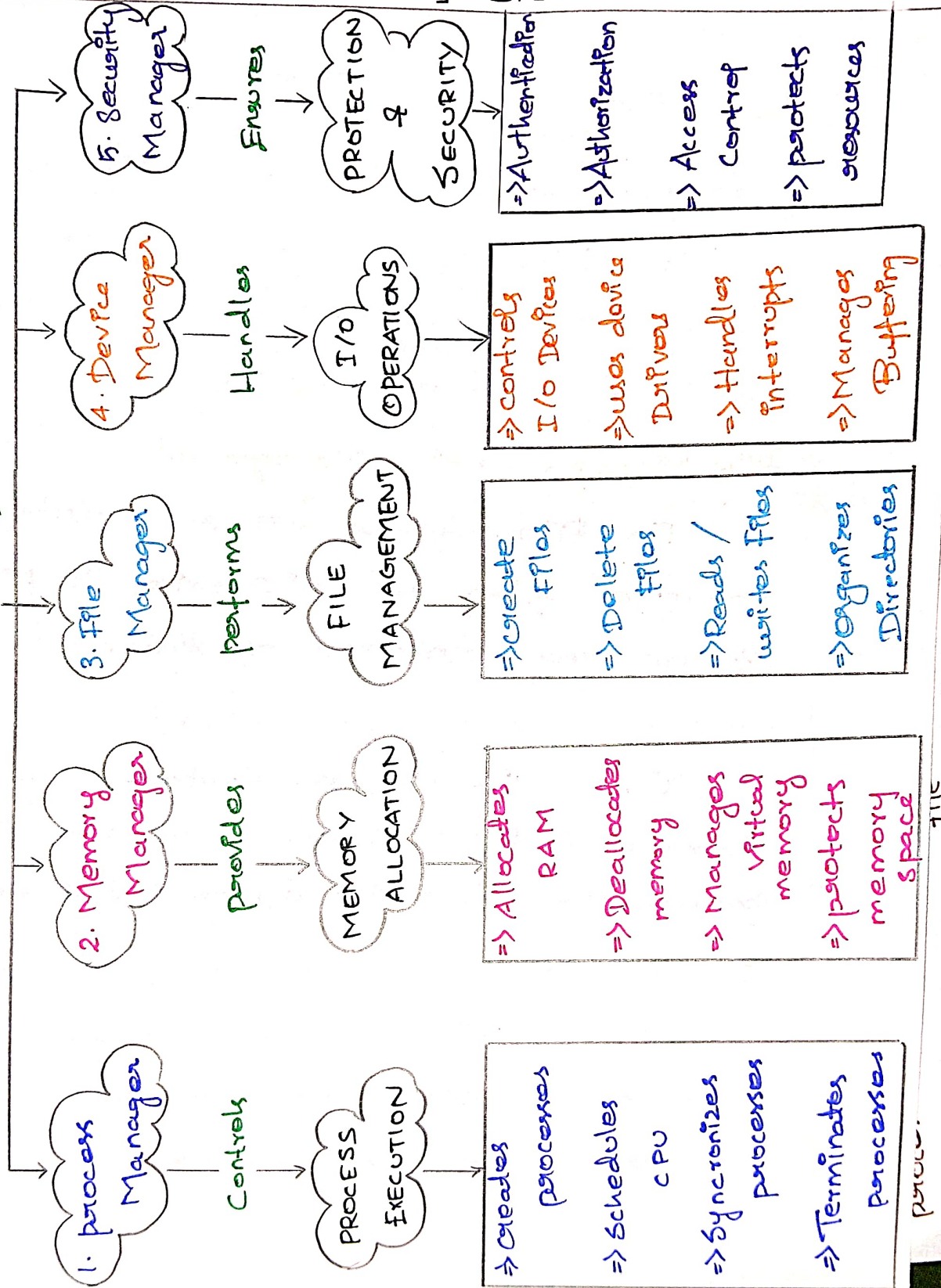
MAPPING

C01 - AT2 => CONCEPT MAPPING

C SA0406 => OPERATING SYSTEMS

1) OPERATING SYSTEMS

Consists of



Relationship . Explanation:

Process Manager $\Rightarrow$ Process Execution

The process manager creates, schedules and terminates processes. It ensures efficient CPU utilization and multitasking.

Memory Manager $\Rightarrow$ Memory Allocation

The Memory manager allocates and deallocates memory for programs. It prevents memory wastage and supports virtual memory.

File Manager $\Rightarrow$ File Management

The file manager creates, deletes and organizes files and directories. It provides secure storage and retrieval of data.

Device Manager $\Rightarrow$ I/O Operations:

The Device manager controls hardware devices through device drivers. It manages communication between the operating system and hardware.

Security Manager $\Rightarrow$ protection & security.

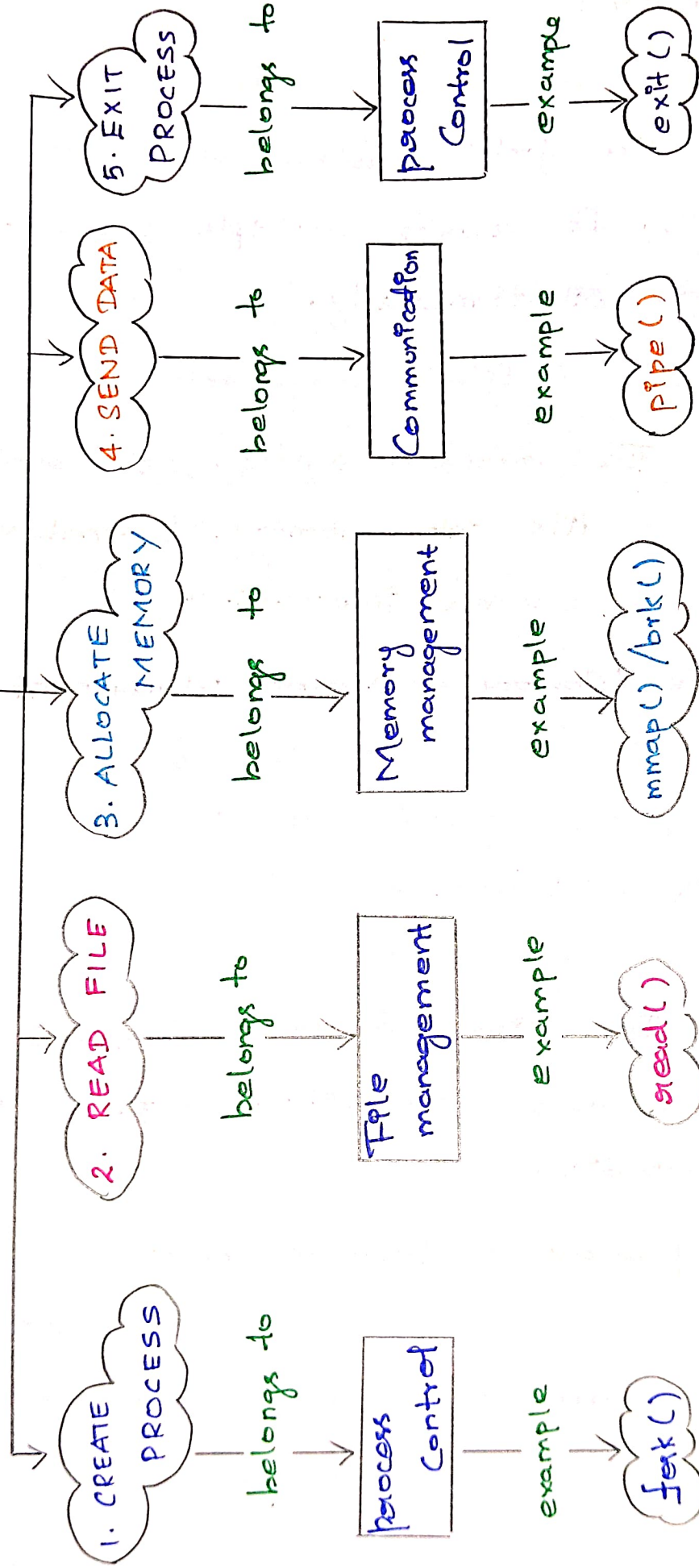
The Security Manager authenticates users, controls permissions and protects system resources from unauthorized access.

2>

APPLICATION PROGRAM

uses

SYSTEM CALLS



Creates a new child process

Reads data from a file

Allocate memory for program

Transfers data between two processes

Terminates the process and frees resources

Explanation :

creates process $\Rightarrow$ process Control

The `fork()` system call creates a new child process. It allows multiple processes to execute simultaneously.

Read File $\Rightarrow$ File Management

The `read()` system call reads data from a file into memory. It enables applications to access stored information.

Allocate Memory $\Rightarrow$ Memory Management :

The `mmap()` or `brk()` system call allocates memory for program execution. It ensures efficient memory utilization.

Send Data $\Rightarrow$ Communication :

The `pipe()` system call transfers data between two processes. It supports inter-process communication.

Exit process $\Rightarrow$ process Control

The `exit()` system call terminates a process and releases all allocated resources.